

MICROCHIP IMPLANTS: Are You Ready For Them?



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Defense Advanced Research Projects Agency (DARPA), an agency of the United States Department of Defense, has developed cyborg insects that can transmit information from sensors implanted into the insect during the pupal stage. The insect's motion is controlled from a micro-electro-mechanical system (MEMS), and it can be used to survey an environment or, say, detect explosives and poisonous gases. Similar high-value and path-breaking work using cutting-edge implantable technology is being done on humans, too

Implants for humans are not new today. Installation of pacemakers and other medical innovations for prosthesis in humans are common and now basic procedures.

Recently, microchip implants were embedded inside the human body, acting as unique lifetime identifiers. A human microchip implant is an integrated circuit (IC) device or radio frequency identification (RFID) transponder encased in a silicate glass and implanted in the human body.

The RFID microchip is basically a tiny, two-way radio, roughly the size of a grain of rice, capable of storing digital information. The sub-dermal implant typically contains a unique 16-digit identification number that can be linked to the information contained in an external database, such as personal identification, law enforcement, medical history, medications, allergies and contact information.

This technology makes it possible, among other things, to instantly verify and confirm the identity of a person.

An RFID implant is capable of holding all the information usually carried on visiting cards. It can transmit this information as one walks through a security checkpoint.

Microchip implants like these have also been used in pets. Some experts have expressed doubts about whether these chips are appropriate for use in humans. However, the fact that these offer many advantages is indisputable.

Background

The first experiments with respect to implants were carried out way back in 1998 by Kevin Warwick, a British scientist. These demonstrated how a computer was able to wirelessly track his movements within the university. He also demonstrated its other applications, such as opening doors and switching on lights.



Fig. 1: A passive RFID Tag (Credit: www.indiamart.com)



Fig. 2: Kevin Warwick was the first to conduct experiments with respect to implants (Credit: www.kevinwarwick.com)